

# Contact-Free Blood Pressure Estimation Based on Pulse Wave Velocity Using mmWave Signals

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**Abstract**—Contact-free blood pressure (BP) estimation leverages the effect of arterial expansion on surrounding mmWave signals to capture pulse waves and subsequently estimate BP. Most current contact-free methods rely on pulse wave analysis, but pulse waves are inherently weak and often lack reliable morphological features. To address this issue, this paper proposes a physiologically interpretable contact-free pulse wave velocity (PWV)-based BP estimation strategy without relying on pulse wave morphological features. Specifically, we develop a PWV-based BP estimation framework, where PWV can be derived by measuring the pulse transit time and pulse transit distance between different body parts using the reflected mmWave signals. To acquire high quality PWV measurement, we exploit the non-uniformity of pulse wave quality in the spatial and frequency domains and the multi-domain characteristics of pulse waves. Based on that, we propose a two-stage candidate space construction strategy to obtain candidate pulse waves from multiple body parts, and a multi-domain filtering and fusion strategy to refine and combine these candidate signals. We design a PWV-based BP estimation system on a 77 GHz mmWave radar and extensively evaluate its performance, with ME  $\pm$  STD of systolic and diastolic BP being  $0.13 \pm 4.87$  mmHg and  $0.33 \pm 4.90$  mmHg, respectively.

**Index Terms**—mmWave signals, contact-free sensing, pulse wave velocity, blood pressure estimation.

## I. INTRODUCTION

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WITH tremendous shifts in lifestyle and global aging population, hypertension has emerged as a major public health challenge, which now affects more than 1.39 billion people worldwide [1]. Hypertension often remains asymptomatic for years yet poses risks for serious complications, making continuous blood pressure (BP) estimation essential for early detection and health maintenance [2]. Currently, most BP estimation methods are contact-based, relying on pulse wave analysis [3]–[5], as pulse waves contain sufficient BP-related information. For example, non-invasive cuff-based measurements and photoplethysmography capture BP-related information using inflatable cuffs or wearable devices. While these methods are widely adopted, they require prolonged skin contact, which can cause discomfort and make them unsuitable for users with sensitive or damaged skin.

Driven by the rapid advancements in wireless sensing technologies [6]–[10], contact-free healthcare has attracted significant attention. To enhance user experience, researchers have developed contact-free BP estimation methods [11]–[14] based on mmWave signals. These methods leverage the unique effects of arterial expansion, induced by blood flow, on surrounding mmWave signals to capture pulse waves in a contact-free manner. Such methods enable BP estimation through pulse wave analysis by replacing the contact-based sensors. Nevertheless, significant challenges still persist in the implementation of contact-free methods for BP estimation. Pulse wave analysis-based methods estimate BP by extracting pulse wave morphological features and therefore heavily rely on complete and clear pulse waves. However, the small amplitude of skin displacement caused by arterial pulsations (only about 0.3 mm [15]) makes it extremely hard to reconstruct complete and clear pulse waves for BP estimation using the reflected mmWave signals. To address this issue, researchers have pioneered the use of deep neural networks to enhance pulse waves and extract BP-related features. Liang et al. [13] designed a pre-training and neural network architecture that extracts BP-related pulse features from sparse and ambiguous mmWave signals. Ran et al. [14] designed a deep learning-based arterial pulse decoder that successfully reconstructs pulse waves from noisy mmWave signals. While the aforementioned methods successfully leverage deep networks to extract BP-related information from noisy mmWave signals, they often sacrifice physiological interpretability and may deviate from established physiological mechanisms. Inspired by the aforementioned works, we pose a naive yet exploratory question: Is it possible to design a contact-free BP estimation system that operates without relying on complete and clear

pulse waves, yet remains physiologically interpretable and user-friendly for daily use?

Related studies have demonstrated that BP, defined as the lateral pressure exerted on arterial walls as pulse waves travel along the arteries, exhibits a strong correlation with pulse wave velocity (PWV) and enable continuous BP estimation via PWV measurements [16]–[18]. Unlike pulse wave analysis methods that depend on morphological features, PWV-based BP estimation methods require the acquisition of pulse transit time (PTT) and pulse transit distance (PTD) between different body parts. Meanwhile, mmWave radar provides favorable spatial resolution capability, enabling simultaneous observation of multiple body parts. Such property inspired this paper to design a contact-free PWV-based BP estimation (PWV-BP) framework using mmWave signals. This framework shifts the focus from reconstructing complete and clear pulse waves to the more feasible task of measuring PWV, while following solid physiological mechanisms and remaining user-friendly for daily use. However, achieving this goal faces several key challenges. Firstly, the lack of existing models for PWV measurement and subsequent BP estimation based on mmWave makes it difficult to ensure the scientific validity of the PWV-based BP estimation strategy. Secondly, the pulse wave is extremely weak and highly susceptible to interference from body movements and environmental noise, making it challenging to obtain PWV by simultaneously measuring pulse waves from multiple body parts using a single mmWave radar.

To address the aforementioned challenges, in this paper, we propose a contact-free PWV-BP framework, which estimates BP by leveraging the well-established physiological relationship between BP and PWV. The proposed PWV-BP framework uses an mmWave radar to simultaneously capture pulse waves from the chest (corresponding to the heart) and the upper arm (corresponding to the brachial artery), from which PWV is derived for BP estimation. Specifically, we firstly construct a human arterial model based on the Fung hyperelastic theory [19], explore the intrinsic relationship between BP and PWV, and validate the reliability of the model using mmWave data. Under the assumption that blood vessels act as distortion free transmission systems, we calculate PWV through PTT and PTD, which can be obtained from the phase difference between the pulse waves at the chest and upper arm simultaneously measured by a single mmWave radar. Secondly, inspired by the non-uniformity of pulse wave quality in both the spatial and frequency domains, we propose a two-stage candidate space construction strategy. This strategy involves adjusting the radar beam orientation in the spatial domain to cover multiple observation windows, and tuning the radar sensing frequency to explore multiple modulation modes of the observed signals, thereby enabling the acquisition of candidate pulse waves from multiple body parts. Additionally, we propose a multi-domain filtering and fusion strategy that leverages frequency sparsity, time stationarity, and spatial aggregation characteristics of pulse waves to refine and combine the candidate signals, enabling accurate extraction of PTT and PTD for PWV calculation. Finally, we design a PWV-BP system using a 77 GHz mmWave radar and comprehensively evaluate its performance under various conditions. Our de-

signed system follows physiological mechanisms, providing a novel and physiologically interpretable solution to contact-free BP estimation.

The main contributions of this paper are summarized as follows.

- 1) We propose a physiologically interpretable, contact-free BP estimation framework based on pulse wave velocity, which operates without relying on pulse wave morphological features.
- 2) We investigate the non-uniformity of pulse wave quality across the spatial and frequency domains, and further analyze the multi-domain characteristics of the pulse waves. Based on these findings, we design a two-stage candidate space construction strategy and a multi-domain filtering and fusion strategy to acquire high quality pulse waves from multiple body parts.
- 3) We develop a PWV-based BP estimation system using a 77 GHz mmWave radar and evaluate its performance under various conditions.

The rest of this paper is organized as follows. Section II reviews related work. Section III defines the problem and introduces the motivation behind our proposed PWV-BP framework. Section IV details the design and architecture of our proposed PWV-BP system. Section V evaluates the PWV-BP system through extensive experiments. Finally, Section VI draws the conclusion.

## II. RELATED WORK

The current gold standard for BP estimation involves the insertion of a catheter into an appropriate artery, such as the radial artery at the wrist, to enable real-time BP monitoring [20], [21]. However, due to its invasive nature, this method requires clinical intervention and increases risks of discomfort and infection. Consequently, it is primarily limited to surgical settings and is impractical for daily home use. In contrast, non-invasive cuff-based methods [22]–[24] are more commonly used in home environments. These methods measure BP indirectly by inflating the cuff to temporarily block blood flow [25]. Nevertheless, they involve significant discomfort due to the occlusion of blood flow and cannot support continuous monitoring.

To improve the comfort and usability of BP monitoring, recent research has increasingly focused on non-occlusive BP estimation methods [26]–[28]. These methods utilize sensors embedded in wearable devices or mobile terminals to collect pulse waves, from which BP is estimated via feature extraction and analysis. Cao et al. [29] developed a wrist-worn BP monitoring device named Crisp-BP, which employed multi-wavelength PPG signals and applied the modified Beer-Lambert law to suppress capillary interference, enabling accurate pulse waveform extraction and subsequent BP estimation. Zhang et al. [30] simultaneously used the microphone and camera of a smartphone to capture heart sounds and fingertip PPG signals, respectively, calculating PTT for BP estimation. Barry et al. [31] proposed a vibrometric force estimation method inspired by cuff-based principles. It used inertial measurement units to quantify finger pressure exerted on the

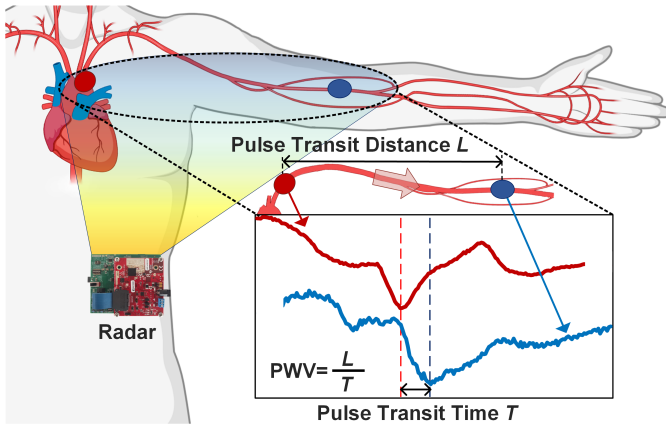


Fig. 1. Principle of PWV measurement based on mmWave radar.

phone, while a camera captured blood volume changes under varying pressure, enabling BP estimation. Despite significant progress in wearable-based BP monitoring, these methods still face limitations such as sensitivity to sensor placement and skin tone, and the potential for skin irritation during long-term wear.

In contrast, non-contact BP monitoring offers improved comfort and convenience, especially for individuals with sensitive skin or for long-term continuous monitoring. Among such methods, vision-based methods estimate BP by detecting skin reflectance changes caused by blood flow. Optical responses from target areas are captured using visual sensors to extract pulse-related fluctuations for BP estimation [32]–[34]. Nevertheless, vision-based methods remain highly sensitive to lighting conditions, skin tone, perspiration, and micro-movements. In addition, their reliance on cameras raises privacy concerns, which limits practical deployment.

To overcome these limitations, an increasing number of researchers have embarked on studying BP estimation using mmWave radar and made significant achievements. Yamaoka et al. [35] experimentally demonstrated the correlation between mmWave echo signal power and BP variations. Kawasaki et al. [36] further explored the feasibility of mmWave-based BP monitoring and successfully extracted BP-related features from the echo signals. However, pulse waves are inherently weak and highly susceptible to environmental interference, which poses significant challenges for mmWave-based BP estimation.

To address these challenges, Shi et al. [11] proposed a novel feature transformation method in the delay-Doppler domain to significantly enhance signal quality for pulse wave construction. Cao et al. [12] introduced a super-resolution spatial scanning technique, which enabled fine-grained scanning of the arm using mmWave radar to extract pulse waves at multiple positions. They also designed a deep recursive convolutional network constrained by BP characteristic transfer functions, which improved the stability of BP estimation. Liang et al. [13] proposed an autocorrelation coefficient-maximizing beamforming method to precisely locate the radial artery and constructed a two-stage neural network architecture for pulse

feature enhancement and BP estimation, thereby boosting estimation accuracy. Ran et al. [14] applied coherent multi-antenna signal fusion to separate spatial sources and employed convolutional neural network-assisted template matching and position tracking to extract pulse waves. A subsequent encoder-decoder network was then used to estimate BP. Hu et al. [37] developed mmFormer, a hybrid Transformer model that combined multi-angle beamforming data augmentation and cross-modal knowledge transfer to achieve highly accurate BP monitoring.

While the aforementioned studies have made valuable contributions to enhancing pulse wave quality and feature extraction, thereby advancing contact-free BP monitoring technologies, most existing methods rely heavily on complex deep learning models to extract pulse-related features from noisy signals. Although these models can achieve effective BP estimation, they often suffer from poor physiological interpretability [38]. Inspired by these efforts, we propose the PWV-BP framework to achieve physiologically interpretable contact-free BP estimation.

### III. PROBLEM DEFINITION AND MOTIVATION

#### A. Problem Definition

In this paper, we aim to achieve physiologically interpretable contact-free BP estimation using a commercial mmWave radar. To this end, we focus on addressing two problems, *i.e.*, how to establish the relationship between PWV measured by mmWave radar and BP, and how to obtain high quality PWV using mmWave radar.

#### B. Motivation

1) *Human Arterial Model*: The propagation of a pulse wave in a human artery can be defined as a disturbance caused by cardiac contraction, propagating along the artery in the form of a wave at a finite velocity, *i.e.*, PWV. The principle of PWV measurement based on mmWave radar is illustrated in Fig. 1. We assume blood vessels as distortion-free transmission systems, such that pulse wave propagation in the artery is considered unaffected by dispersion or waveform distortion, thereby simplifying the calculation of PWV. Accordingly, PWV can be calculated from the PTD and the PTT between two body parts, as follows

$$PWV = \frac{L}{\frac{1}{N_t} \sum_{n=1}^{N_t} T(n)}, \quad (1)$$

where  $L$  and  $T$  denote the PTD and PTT between the chest and upper arm, respectively.  $N_t$  represents the number of pulse wave intervals (PWI) within the measurement period. In our proposed PWV-BP framework, PTT and PTD are obtained by simultaneously measuring the phase and position differences of pulse waves at the two parts using a single mmWave radar, as shown in Fig. 1.

For a long and straight artery filled with incompressible and non-viscous blood, Fung's hyperelastic theory [19] gives the relationship between PWV and BP, as follows

$$BP = aPWV^2 + b, \quad (2)$$



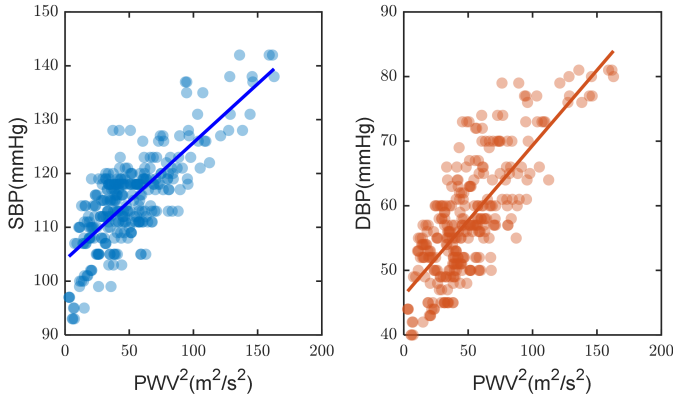


Fig. 2. Relationship between  $PWV^2$  and BP.

where  $a$  is a scaling coefficient and  $b$  is an additive offset. Both  $a$  and  $b$  are insensitive to BP, as they depend primarily on the material properties of the blood vessel and the geometry of the artery. Hence, they can be approximated as constants and determined experimentally.

To intuitively illustrate the relationship between PWV measured by mmWave radar and actual BP, we fit  $PWV^2$  to systolic BP (SBP) and diastolic BP (DBP) using the least squares method, as shown in Fig. 2. For the same user, BP is positively correlated with  $PWV^2$ , satisfying the mapping relationship described in Eq. (2). The results validate the feasibility of estimating BP using radar-measured PWV.

Although the points in Fig. 2 are uniformly distributed around the fitted line, they still exhibit a certain degree of dispersion. To enhance the accuracy of BP estimation, additional physiological factors affecting BP should be considered. For example, an increase in heart rate leads to an increase in cardiac output. If arteries are assumed to be purely resistive, BP would increase linearly with heart rate [5]. Therefore, we establish a human arterial model as follows

$$\begin{aligned} SBP &= a_s \cdot PWV^2 + \beta_s^T \cdot \mathbf{x} + b_s, \\ DBP &= a_d \cdot PWV^2 + \beta_d^T \cdot \mathbf{x} + b_d, \end{aligned} \quad (3)$$

where  $\mathbf{x}$  represents additional features related to BP, *i.e.*, statistical features of PTT, heart rate, *etc.*, and  $\{a_s, a_d, \beta_s, \beta_d, b_s, b_d\}$  are parameters to be optimized. By incorporating these physiological features, our proposed human arterial model offers a more comprehensive and physiologically realistic representation of BP.

2) *Non-Uniformity in the Quality of Pulse Waves*: To measure PWV using radar, it is necessary to simultaneously monitor pulse waves at both the chest and upper arm using a single radar device, as shown in Fig. 1. The pulse wave at the chest, *i.e.*, the heartbeat, is weak, with an amplitude of approximately 0.5 mm [39], and is highly susceptible to respiratory interference. This presents a typical challenge of weak signals and severe interference. In comparison, the pulse wave at the upper arm is even weaker, with an amplitude of less than 0.03 mm, and originates from a smaller target region, as the arterial diameter is only approximately 3 mm [40]. This presents a typical challenge of weak signals and small targets.

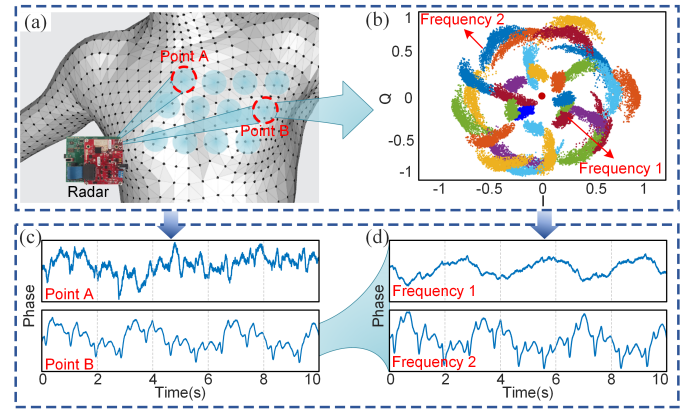


Fig. 3. Non-uniformity in the quality of pulse waves. (a) In spatial domain. (b) In frequency domain. (c) Phase extracted from two spatial points. (d) Phase extracted from two sensing frequencies.

To address the challenges of small targets, weak signals, and severe interference in pulse wave sensing, we conduct a systematic exploration into how different sensing spatial positions and sensing frequencies affect the ability of radar to sense pulse waves. We observe that the signal quality exhibits significant non-uniformity across both the spatial and frequency domains, as shown in Fig. 3. These findings offer important insights for signal enhancement and reliable PWV measurement. In the spatial domain, the modulation effect of the cardiovascular system on different body points varies depending on their proximity to the heart or artery [41], [42]. As shown in Fig. 3(c), the received signals at different positions exhibit significant differences: point B shows a higher quality and clearer waveform compared to point A. This confirms the spatial-dependent non-uniformity in pulse wave quality. In the frequency domain, pulse waves exhibit different modulation effects across various radar sensing frequencies [43]. As illustrated in Fig. 3(b), the clusters in the In-phase and Quadrature (IQ) domain represent the received signals corresponding to different sensing frequencies. In the IQ domain, clusters near the center, such as those from frequency 1, exhibit low energy and limited variation, while clusters farther from the center, such as those from frequency 2, are characterized by higher energy and more distinct variation. Meanwhile, as illustrated in Fig. 3(d), signals from frequency 1 are only meaningless noise, while signals from frequency 2 exhibit distinct pulse wave characteristics. These observations confirm the frequency-dependent non-uniformity in pulse wave quality.

Based on the above observations, we propose a two-stage candidate space construction strategy to address the challenges of small targets, weak signals, and severe interference in pulse wave sensing. In the spatial domain construction stage, the radar adjusts its observation points to ensure sufficient coverage of the target region. In the frequency domain construction stage, the radar tunes its sensing frequency to alter the modulation characteristics of the received signals, thereby changing the interaction between pulse waves and interference. This strategy increases the probability of capturing high quality



pulse waves from multiple body parts for PWV measurement, making it feasible to realize contact-free BP estimation in daily scenarios using a commercial mmWave radar.

#### IV. SYSTEM DESIGN

##### A. System Architecture

The system architecture of our proposed PWV-BP system, illustrated in Fig. 4, consists of four modules: mmWave radar sensing module, two-stage candidate space construction module, signal filtering and fusion module, and BP estimation module.

The mmWave radar sensing module collects signals reflected from the human body and separates them into different spatial points through beamforming, thereby providing measurement information from various body parts to the subsequent module. The two-stage candidate space construction module adjusts signal quality by applying spatial domain candidate space construction scheme and frequency domain candidate space construction scheme, thereby measuring candidate pulse waves. The signal filtering and fusion module further refines and combines these signals based on the frequency sparsity, time stationarity, and spatial aggregation characteristics of pulse waves, extracting the pulse wave position (PWP) and PWI sequence. The BP estimation module calculates PTT and PTD from the extracted PWP and PWI sequence, derives PWV, and ultimately estimates BP through our proposed human arterial model.

##### B. mmWave Radar Sensing

The radar sensor used for BP estimation is equipped with a multi-antenna array. It transmits a frequency-modulated continuous wave signal  $S_{TX}$  as follows

$$S_{TX}(t) = G e^{j\theta_0} e^{j(2\pi f_0 t + \pi k t^2)}, \quad (4)$$

where  $t$ ,  $G$ ,  $\theta_0$ ,  $f_0$ , and  $k$  represent the fast time, amplitude, initial phase, starting frequency, and the frequency modulation slope, respectively.

When the transmitted signal encounters a human body, it will generate the reflected signal  $S_{RX}$ , which can be received by the radar as

$$S_{RX}(t) = \sigma G e^{j\theta_0} e^{j(2\pi f_0(t - \frac{d}{c}) + \pi k(t - \frac{d}{c})^2)}, \quad (5)$$

where  $\sigma$  denotes the attenuation coefficient,  $d$  represents the round-trip distance between the human body and the radar, and  $c$  indicates the speed of light. The radar mixes the received signal  $S_{RX}$  with the transmitted signal  $S_{TX}$  to obtain the intermediate frequency (IF) signal  $S_{IF}$  defined by

$$S_{IF}(t) = S_{TX}(t) S_{RX}^*(t) \approx \bar{G} e^{j2\pi(f_0 + kt)\frac{d}{c}}, \quad (6)$$

where  $\bar{G} = \sigma G^2$  denotes the IF signal amplitude and  $S_{RX}^*(t)$  indicates the complex conjugate of  $S_{RX}(t)$ .

Each voxel on the human body can be uniquely identified by its polar coordinates  $(r, \theta)$  from the radar as the origin, where a voxel denotes the smallest indivisible spatial unit of the human body. For a specific voxel, the slight differences in propagation

paths to different antennas are projected as phase variations in the received IF signals. The received IF signals from the virtual antenna array are phase-compensated and coherently combined to obtain the received signal of voxel  $(r, \theta)$ , which is denoted as

$$S_{r,\theta}(t, \bar{t}) = \sum_{n=1}^{N_a} \bar{G}(\bar{t}) e^{j2\pi(f_0 + kt)\frac{d_{r,\theta}(n, \bar{t})}{c}}, \quad (7)$$

where  $\bar{t}$  represents the frame time,  $N_a$  is the number of virtual channels formed by the pairs of transmitting and receiving antennas,  $d_{r,\theta}(n, \bar{t})$  denotes the round-trip distance from the voxel  $(r, \theta)$  to the transmit-receive antenna pair in the  $n$ -th channel at frame  $\bar{t}$ .

After applying digital beamforming, we can steer the radar beam to capture reflections from different points on the human body, where a point corresponds to the minimal resolvable spatial unit in radar sensing. Due to the inherent beamwidth limitations, each point encompasses multiple voxels, resulting in the beamformed signal that is a superposition of reflections from these voxels [43]. Therefore, the beamformed signal at a given point  $(r, \theta)$  can be expressed as

$$S_{r,\theta}(t, \bar{t}) = \sum_{n=1}^{N_a} \sum_{m=-M_1}^{M_2} \bar{G}(\bar{t}) e^{j2\pi(f_0 + kt)\frac{d_{r,\theta}(n, \bar{t}) + 2l \sin \theta \cdot m}{c}}, \quad (8)$$

where  $M_1, M_2 \geq 0$ , each point contains  $M_1 + M_2 + 1$  voxels, and  $l$  denotes the spacing distance between adjacent voxels.

##### C. Two-Stage Candidate Space Construction Strategy

1) *Spatial Domain Candidate Space Construction*: The received IF signal contains both dynamic energy induced by body motion and static energy generated by the torso. To highlight pulse waves, we utilize an energy accumulation-based pulse wave localization mechanism [44]. Firstly, by performing a fast Fourier transform (FFT) along the fast-time dimension  $t$  on the beamformed signal  $S_{r,\theta}(t, \bar{t})$ , a time-series signal  $\tilde{S}_{r,\theta}(\bar{t})$  along the frame sequence can be formed for each point  $(r, \theta)$ . By extracting its phase, we obtain the motion waveform perceived by the radar corresponding to each point. It is noteworthy that although all components of the signal can be used to extract the pulse waveform, phase information is more suitable for detecting small scale vibration [45]. Secondly, apart from static interference caused by the torso, human motion primarily includes pulse waves (0.67-3.67 Hz), respiration (0.2-0.67 Hz), and random movements. To amplify pulse wave energy, we accumulate the signal energy within the pulse wave frequency band to obtain the pulse wave energy spectrum, as shown in Fig. 5(a). This spectrum effectively reveals the distribution of pulse wave energy across different body parts, while mitigating the effects of static interference, respiration, and random noise. Finally, pulse waves originating from the heart and the brachial artery, which is close to the skin, exhibit stronger amplitudes and can be more easily detected by radar. Using the constant false alarm rate detection (CFAR) algorithm [46], we identify the positions of the chest and upper arm, denoted as  $(r^1, \theta^1)$  and

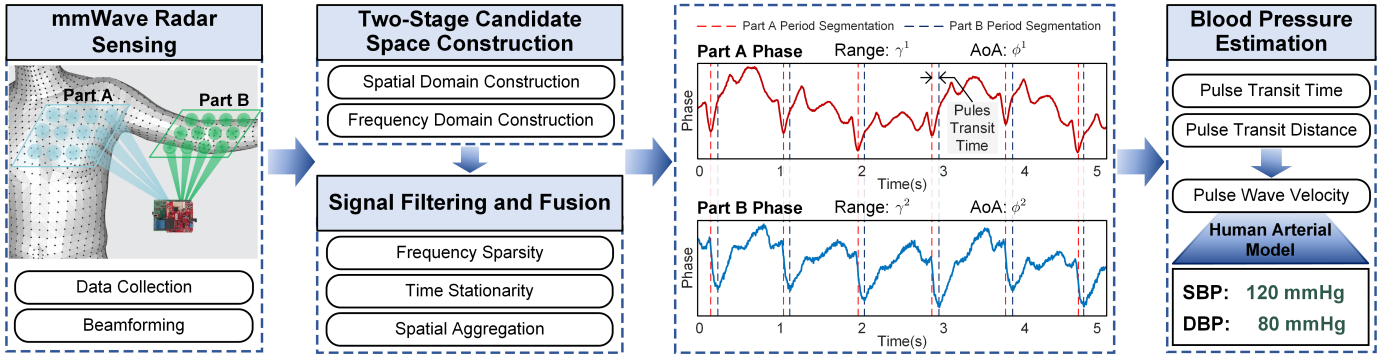


Fig. 4. System architecture of our proposed PWV-BP system.

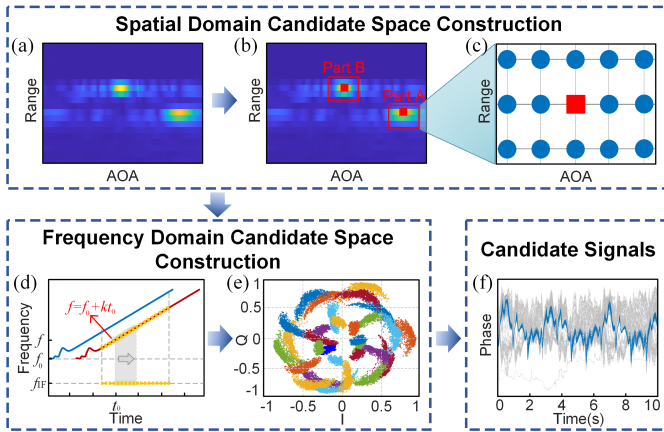


Fig. 5. Two-stage candidate space construction strategy. (a) Pulse wave energy spectrum. (b) The results of CFAR. (c) Selected spatial points within one of the parts. (d) Construction of different sensing frequencies. (e) Raw signals constructed in the complex plane at different sensing frequencies. (f) Candidate signals.

$(r^2, \theta^2)$ , respectively, *i.e.*, the strongest energy points in Part A and Part B of Fig. 5(b), respectively.

While the pulse wave energy spectrum reflects the frequency characteristics of each point, it does not fully represent the signal quality of pulse waves. To comprehensively capture high quality pulse waves, we construct two point clouds centered at  $(r^i, \theta^i)$  to cover the chest and upper arm, as shown in Fig. 5(c). Each point cloud  $\mathcal{P}^i$  ( $i = 1, 2$  correspond to the chest and upper arm, respectively) consists of 15 spatially distributed points, denoted as

$$\begin{aligned} \mathcal{P}^i &= \{(r, \theta) | r \in \Gamma^i, \theta \in \Theta^i(r)\}, \\ \text{s.t. } \Gamma^i &= r^i \pm p\Delta d, \\ \Theta^i(r) &= \arg \tan \left( \frac{r \sin \theta^i \pm qh}{r \cos \theta^i} \right), \end{aligned} \quad (9)$$

where  $p = 0, 1$ ,  $q = 0, 1, 2$ ,  $\Delta d$  denotes the range resolution, and  $h$  represents the predefined interval between adjacent points at the same range, which is set to 0.1 m.

## 2) Frequency Domain Candidate Space Construction:

After the spatial candidate points are constructed, candidate

sensing frequencies are further generated for each point, because the signal quality at a given spatial position can also vary with the radar sensing frequency. To investigate the impact of the radar sensing frequency on pulse wave quality, we begin with a simplified scenario. Specifically, we assume that point  $(r', \theta')$  contains only two voxels, and the difference in round-trip distance between these two voxels and the radar is  $D = 2l \sin \theta$ . Under this condition, the received signal at this point can be simplified from Eq. (8) to

$$S_{r', \theta'}(t, \bar{t}) = \sum_{n=1}^{N_a} \bar{G}(\bar{t}) e^{j2\pi F \frac{d_{r', \theta'}(n, \bar{t})}{c}} [1 + e^{j2\pi \frac{FD}{c}}], \quad (10)$$

where  $F = f_0 + kt$  involves frequency-related components. When  $F = c/2D$ , the signals reflected from the two voxels destructively interfere with each other, resulting in the point signal  $S_{r', \theta'}(t, \bar{t}) = 0$ . In contrast, when  $F = c/D$ , the received signal amplitude doubles due to enhancement between the two voxel signals. These results suggest that the superposition of voxel signals within a point is strongly influenced by the radar sensing frequency, which can be tuned to alter signal quality.

Without changing the radar hardware parameter configuration, we segment the beamformed signal using a sliding window mechanism to obtain received signals with different radar sensing frequencies, as illustrated in Fig. 5(d). Specifically, for a sliding window with a length of  $w$  and a step of  $\Delta t$ , the beamformed signal can be divided into multiple sub-signals, where the  $i$ -th sub-signal can be expressed as

$$S_{r, \theta, f}(:, \bar{t}) = S_{r, \theta}((i-1)\Delta t + 1 : (i-1)\Delta t + w, \bar{t}). \quad (11)$$

The radar sensing frequency of the  $i$ -th sub-signal is defined by  $f = f_0 + k(i-1)\Delta t/\epsilon$ , where  $\epsilon$  denotes the sample rate. Subsequently, performing an FFT along the fast-time dimension on the sub-signal forms a time-series signal  $\bar{S}_{r, \theta, f}(\bar{t})$  at each point  $(r, \theta)$  and for each sensing frequency  $f$ , along the frame sequence. Fig. 5(e) illustrates the received signals in the IQ domain at a fixed spatial position for various sensing frequencies. Evidently, the sensing frequency significantly influences the IQ distribution of the received signals, resulting in distinct patterns even at the same position.

In summary, we first apply a spatial domain candidate space

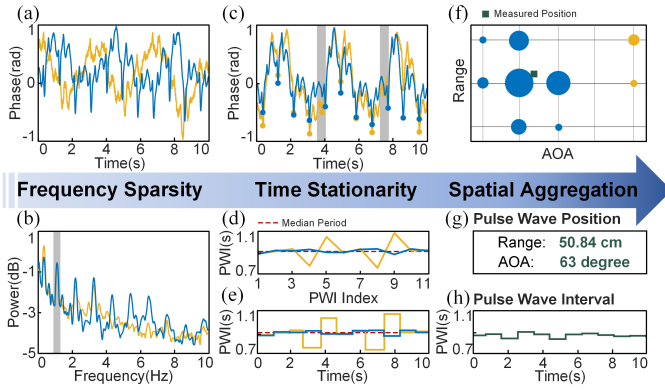


Fig. 6. Multi-domain characteristics of pulse waves. (a) The phase of two candidate signals. (b) The spectrum of the two candidate signals in (a). (c) Pulse wave period segmentation. (d) PWI sequence derived from period segmentation. (e) Alignment of the PWI sequence with the time axis. (f) Pulse waves distribution in the space. (g) The measured pulse wave position. (h) The measured pulse wave interval sequence.

construction scheme to adjust the observation position of the radar, followed by a frequency domain candidate space construction scheme to further refine the radar sensing frequency. As a result, candidate signals from the chest and upper arm are obtained, denoted as  $\bar{S}_{\mathcal{P}^1,f}(\bar{t})$  and  $\bar{S}_{\mathcal{P}^2,f}(\bar{t})$ , respectively. Each candidate signal carries three distinct attributes, namely range  $r$ , angle  $\theta$ , and sensing frequency  $f$ . Fig. 5(f) shows all candidate signals, where blue indicates high quality signals and gray indicates low quality ones. It should be noted that the spatial scanning range is intentionally set larger than the actual signal region to ensure full coverage of the target areas, which inevitably introduces low quality signals. Moreover, variations in radar sensing frequency may either enhance or degrade the signal quality, resulting in candidate signals with different quality levels. To acquire high quality signals, we propose a multi-domain filtering and fusion strategy in the next section.

#### D. Multi-Domain Filtering and Fusion Strategy

Selecting high quality pulse waves from the candidate signals  $\bar{S}_{\mathcal{P}^i,f}(\bar{t})$  at the chest and upper arm remains a challenging task. Therefore, this section proposes a multi-domain filtering and fusion strategy that fully leverages the multi-domain characteristics of pulse waves, *i.e.*, frequency sparsity, time stationarity, and spatial aggregation, as illustrated in Fig. 6 to sequentially refine and combine the pulse waves across the frequency, time, and spatial domains, so as to ultimately measure the PWPs and PWI sequence accurately.

1) *Frequency Sparsity*: High quality candidate signal shows clear pulse wave periodicity (blue), whereas low quality one (yellow) is primarily dominated by low-frequency interference such as respiration and meaningless high-frequency noise, as illustrated in Fig. 6(a). According to the time-frequency duality principle, stronger periodicity in the time domain corresponds to more pronounced sparsity in the frequency domain. Therefore, we evaluate the periodicity of the candidate signals by leveraging frequency sparsity. We perform an FFT

along the frame-time dimension on the phase of the candidate signals to obtain their frequency domain representation  $\vec{S}_{\mathcal{P}^i,f}$ , as shown in Fig. 6(b). Due to the rich harmonic components of pulse waves, the high quality blue signal exhibits pronounced sparsity in both the fundamental pulse frequency band and its harmonics. In contrast, the yellow signal shows a single peak in the low-frequency respiratory band and appears relatively flat in the pulse frequency band and its harmonics, indicating a lack of sparsity. Subsequently, we assess frequency sparsity by measuring the peak-to-peak amplitude of fundamental frequency of the pulse wave. Specifically, we first employ a peak detection algorithm [47] to identify the index  $\hat{k}_{\mathcal{P}^i,f}$  corresponding to the fundamental frequency of the pulse wave within its typical fluctuation interval  $[0.8, 2]$  Hz, as well as the index  $\check{k}_{\mathcal{P}^i,f}$  of the valley adjacent to the peak at index  $\hat{k}_{\mathcal{P}^i,f}$ . Then, we compute the peak-to-peak value, as follows

$$K_{\mathcal{P}^i,f} = \vec{S}_{\mathcal{P}^i,f}(\hat{k}_{\mathcal{P}^i,f}) - \vec{S}_{\mathcal{P}^i,f}(\check{k}_{\mathcal{P}^i,f}). \quad (12)$$

Finally, we use the peak-to-peak value  $K_{\mathcal{P}^i,f}$  to quantify the frequency sparsity of the candidate signals. The larger the peak-to-peak value, the more significant the sparsity, and the higher the signal quality. By applying frequency sparsity, we effectively filter out low quality candidate signals generated by the torso near the chest and upper arm, and denote the retained signal as  $\bar{S}_{\mathcal{P}^i,f}(\bar{t})$ .

2) *Time Stationarity*: To obtain the PTT, the pulse waveform must be segmented based on its periodicity. However, high-frequency noise can distort the waveform and cause peak shifts, as illustrated in Fig. 6(c). The yellow waveform shows noticeable peak shifts within the shaded areas, which can introduce errors in pulse period calculation and consequently affect the estimation of PWV and BP. These shifts caused by noise introduce noticeable fluctuations in PWI, as shown in Fig. 6(d). When a peak shifts forward, the preceding pulse period becomes shorter while the following one becomes longer. Conversely, a backward shift results in the opposite effect. In contrast, high quality pulse waveforms are less affected by noise and exhibit more stable and realistic pulse periods, whose PWI variations are mainly caused by physiological heart rate variability. Therefore, we evaluate the quality of the candidate signals by leveraging time stationarity, and subsequently derive PWI sequence based on high quality signals.

Specifically, we first apply a peak detection algorithm [47] to obtain the set of local peak indices for each signal in candidate signals  $\bar{S}_{\mathcal{P}^i,f}(\bar{t})$ , denoted as  $\{pk_1, pk_2, \dots, pk_j, \dots\}$ . Then, the  $j$ -th PWI of this signal is defined as the time difference between two consecutive pulse wave peaks, as follows

$$\psi_j = (pk_{j+1} - pk_j)\tau, \quad (13)$$

where  $\tau$  is the frame periodicity. The PWI sequence is expressed as  $I = [\psi_1, \psi_2, \dots, \psi_j, \dots]$ . Next, we align the PWI sequence with the time axis, as illustrated in Fig. 6(e), where the length of each line segment represents the duration of a PWI, and the vertical position corresponds to the magnitude



of its respective interval. Finally, we compute the variance  $var$  of each PWI sequence to quantify its stability, as follows

$$var = \frac{1}{J} \sum_{j=1}^J \left( I(j) - \frac{\sum_{j=1}^J I(j)}{J} \right)^2, \quad (14)$$

where  $J$  denotes the number of PWIs. Based on time stationarity, we retain the candidate signals with lower PWI variance and filter out the low quality signals with higher variance.

3) *Spatial Aggregation*: The candidate signals retained after time and frequency domain filtering all exhibit clear pulse wave periodicity. In this section, we further filter out spatial outliers and fuse the remaining signals based on their spatial aggregation. Their spatial distribution is illustrated in Fig. 6(f), where the size of each circle indicates the number of candidate signals obtained from different radar sensing frequencies at that position. Blue regions represent aggregate candidate signals, while yellow regions indicate sparse ones. Due to the propagation of pulse waves along arteries with a certain delay between distant spatial points, directly combining candidate signals may introduce inaccuracies in the PWI sequence. Therefore, we utilize spatial aggregation to determine the weights of the candidate signals, and then compute the final PWP and PWI sequence at the chest and upper arm through a weighted combination.

Specifically, we first apply the DBSCAN algorithm [48] to cluster the spatial positions of the candidate signals and remove spatial outliers, *i.e.*, the yellow regions, to eliminate the interference they introduce. Next, we compute the spatial weight for each of the remaining candidate signals, as follows

$$\xi_{r_n, \theta_n}^i = \frac{N_{r_n, \theta_n}^i}{N_s^i}, \quad 1 \leq n \leq N_s^i, \quad (15)$$

where  $N_{r_n, \theta_n}^i$  denotes the number of candidate signals located at  $(r_n, \theta_n)$ ,  $\xi_{r_n, \theta_n}^i$  represents the weight of the  $n$ -th candidate signal at the chest or upper arm, and  $N_s^i$  denotes the total number of candidate signals after clustering. Finally, we denote the PWP and PWI sequence of the  $n$ -th clustered candidate signal from the chest or upper arm as  $(r_n^i, \theta_n^i)$  and  $I_{r_n, \theta_n}^i$ , respectively. The spatial positions of these candidate signals are then fused through a weighted combination to obtain  $P^i(\gamma^i, \phi^i)$ , where

$$\begin{aligned} \gamma^i &= \sum_{n=1}^{N_s^i} \xi_{r_n, \theta_n}^i \cdot r_n^i, \\ \phi^i &= \sum_{n=1}^{N_s^i} \xi_{r_n, \theta_n}^i \cdot \theta_n^i. \end{aligned} \quad (16)$$

We define  $P^i$  as the PWP at the chest or upper arm. Similarly, the PWI sequences of different candidate signals are combined through a weighted combination as

$$\mathcal{I}^i = \sum_{n=1}^{N_s^i} \xi_{r_n, \theta_n}^i \cdot I_{r_n, \theta_n}^i, \quad (17)$$

where  $\mathcal{I}^i = [\psi_1^i, \psi_2^i, \dots, \psi_j^i, \dots]$  denotes the PWI sequence at the chest or upper arm. Through frequency sparsity, time

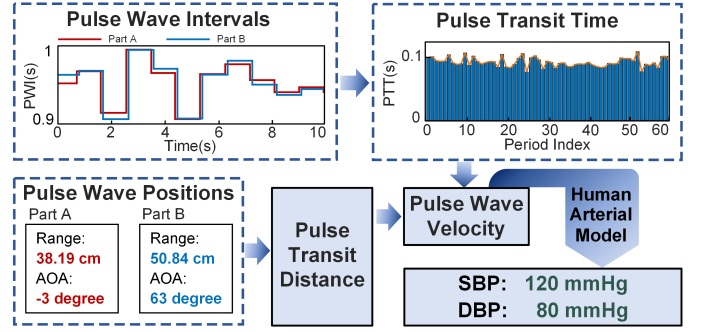


Fig. 7. The working principle of the BP estimation.

stationarity, and spatial aggregation, we effectively refine and combine the candidate signals, resulting in a stable PWP and PWI sequence.

### E. Blood Pressure Estimation

Through the above procedures, the PWP and PWI sequences at the chest and upper arm are obtained. Based on these measurements, this section further estimates BP, as shown in Fig. 7. As depicted in Fig. 1, the PTD can be approximated by the spatial distance between the chest and upper arm, as follows

$$L = \gamma^1 \sin \phi^1 - \gamma^2 \sin \phi^2, \quad (18)$$

where  $\gamma^1$  and  $\phi^1$  denote the range and angle of the chest, respectively, and  $\gamma^2$  and  $\phi^2$  denote the range and angle of the upper arm, respectively. Based on the PWI sequences at the chest  $\mathcal{I}^1$  and upper arm  $\mathcal{I}^2$ , the PTT sequence is defined as the cumulative time difference between them, expressed as

$$T(j) = \sum_{n=1}^j (\mathcal{I}^1(n) - \mathcal{I}^2(n)), \quad (19)$$

where  $\mathcal{I}^i(n)$  denotes the  $n$ -th PWI at the chest or upper arm. By substituting the PTD  $L$  and PTT  $T$  into Eq. (1), the PWV can be obtained.

We further estimate BP from PWV based on our proposed human arterial model (Eq. (3)). To obtain the model parameters, we formulate the objective function for BP estimation, as follows

$$\begin{aligned} \mathbf{F} &= \arg \min_{\mathbf{F} \in \mathbb{R}^{16}} \sum_{j=1}^{\mathcal{M}} \left[ (\hat{\text{SBP}}_{(j)} - \text{SBP}_{(j)})^2 + (\hat{\text{DBP}}_{(j)} - \text{DBP}_{(j)})^2 \right] \\ &\quad + \lambda (a_s^2 + a_d^2 + \|\beta_s\|_2^2 + \|\beta_d\|_2^2), \\ \text{s.t.} \quad &\hat{\text{SBP}}_{(j)} = a_s \cdot \text{PWV}_{(j)}^2 + \beta_s^\top \cdot \mathbf{x} + b_s, \\ &\hat{\text{DBP}}_{(j)} = a_d \cdot \text{PWV}_{(j)}^2 + \beta_d^\top \cdot \mathbf{x} + b_d, \end{aligned} \quad (20)$$

where SBP and  $\hat{\text{SBP}}$  denote the ground-truth and estimated values of SBP, respectively, and DBP and  $\hat{\text{DBP}}$  denote the ground-truth and estimated values of DBP, respectively.  $\mathcal{M}$  denotes the number of BP samples.  $\mathbf{x}$  consists of six BP-related features, including the mean, variance, trend, and

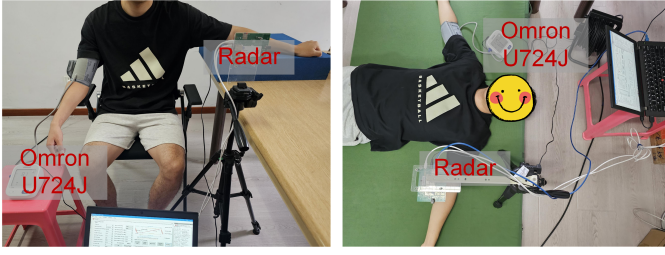


Fig. 8. The experimental scenarios for sitting and lying.

interquartile range of PTT, as well as heart rate and PWV.  $\lambda$  denotes the regularization coefficient, the parameter set is defined as  $\mathbf{F} = \{a_s, \beta_s, b_s, a_d, \beta_d, b_d\} \in \mathbb{R}^{16}$ , and  $\|\cdot\|_2$  represents the  $\ell_2$ -norm. To determine the optimal parameter set, the defined objective function is optimized to minimize the estimation error between the predicted and actual BP values. We employ a numerical optimization approach to obtain the optimal parameters, ensuring convergence to a stable solution. Subsequently, the obtained optimal parameter set is substituted back into Eq. (3) to calculate the final BP estimation results.

## V. EXPERIMENTAL EVALUATIONS

### A. Experimental Setup and Quantitative Evaluation Metrics

We design a PWV-based BP estimation system using the 77 GHz Texas Instruments AWR1443BOOST mmWave radar and comprehensively evaluate its performance under various conditions. Table I provides a detailed overview of the radar configuration. We use a Texas Instruments DCA1000EVM board to collect raw mmWave signals, which are then transferred to a laptop via Ethernet cable and further processed in MATLAB.

TABLE I. Default parameter settings of the mmWave radar.

| Parameters         | Values    | Parameters       | Values   |
|--------------------|-----------|------------------|----------|
| Starting Frequency | 77 GHz    | Number of TX     | 3        |
| Frequency Slope    | 75 MHz/us | Number of RX     | 4        |
| Sample Rate        | 2.8 Mps   | Maximum Range    | 2.8 m    |
| Bandwidth          | 3.43 GHz  | Range Resolution | 0.044 m  |
| Number of Samples  | 128       | Frame Time       | 5 ms     |
| Number of Chirps   | 5         | AoA Resolution   | 0.25 rad |

All experiments are conducted in quiet indoor environments, with room temperatures ranging from 18 °C to 28 °C. The study involves 9 users, including 6 males and 3 females, across 5 different scenarios. Users remain relatively still during the experiments and are allowed to breathe normally. As illustrated in Fig. 8, each user is instructed to sit on a chair with the left arm resting on a table, or to lie flat with the left arm extended perpendicular to the torso, ensuring that the heart and brachial artery are aligned on the same horizontal plane. An mmWave radar is mounted 30–40 cm in front of the upper arm, and the corresponding lateral resolution is approximately 7.5–10 cm. During data collection, users remain motionless, and the recorded data are organized into 20s segments for subsequent analysis. At the same time as the radar collection, a FDA-approved arm-cuff BP device (Omron U724J) is used to obtain

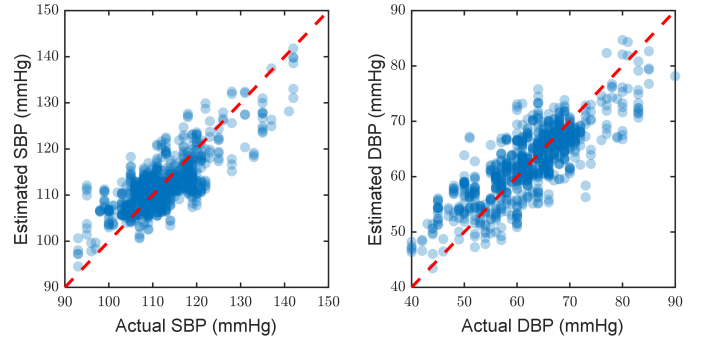


Fig. 9. Correlation between PWV-BP estimations and actual BP.

ground-truth measurements. The cuff is worn on the right arm at heart level to ensure accuracy. Data are collected on different days and at different times, with a minimum interval of 2 minutes between recordings. Across different recording sessions, BP is modulated through deep breathing, breath-holding, and squat tasks. The entire data collection process spans 6 months.

In the following experiments, we evaluate the effectiveness of our proposed system employing three common metrics, *i.e.*, mean absolute error (MAE), mean error (ME), and standard deviation of mean error (STD), as follows

$$\begin{aligned}
 \text{MAE} &= \frac{\sum_{n=1}^N |\hat{\text{BP}}_n - \text{BP}_n|}{N}, \\
 \text{ME} &= \frac{\sum_{n=1}^N (\hat{\text{BP}}_n - \text{BP}_n)}{N}, \\
 \text{STD} &= \sqrt{\frac{\sum_{n=1}^N (\hat{\text{BP}}_n - \text{BP}_n - \text{ME})^2}{N}},
 \end{aligned} \tag{21}$$

where  $\hat{\text{BP}}_n$  denotes the estimated BP of the  $n$ -th sample,  $\text{BP}_n$  represents the ground truth, and  $N$  indicates the total number of samples.

### B. Overall Performance Analysis

To evaluate the effectiveness of PWV-BP, we perform both qualitative and quantitative analyses.

1) *Qualitative Assessment:* The relationship between the BP estimated by the PWV-BP and actual BP is shown in Fig. 9. All data points exhibit a clear positive correlation along the reference line, with SBP ranging from 93 mmHg to 142 mmHg and DBP ranging from 40 mmHg to 90 mmHg. In addition, the points are tightly clustered around the reference line with low dispersion. These results demonstrate that the BP estimated by the PWV-BP agrees well with the actual BP and reflects the true BP level across the full measurement range.

We present the Bland-Altman plots of the SBP and DBP estimated by the PWV-BP in Fig. 10. The x-axis represents the actual BP, while the y-axis shows the PWV-BP estimation errors. The red line denotes the ME, and the black lines denote the limits of agreement (LOA) [49], defined as  $\text{ME} \pm 1.96 \times \text{STD}$ . The results indicate that the ME and LOA for SBP are 0.13 mmHg and -9.42 to 9.68 mmHg, respectively,

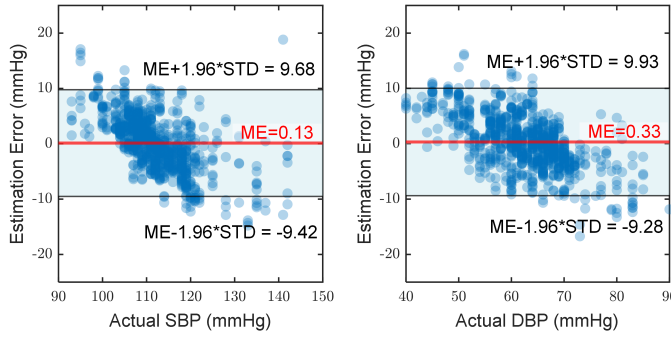


Fig. 10. Bland-Altman plots of BP estimation errors based on PWV-BP.

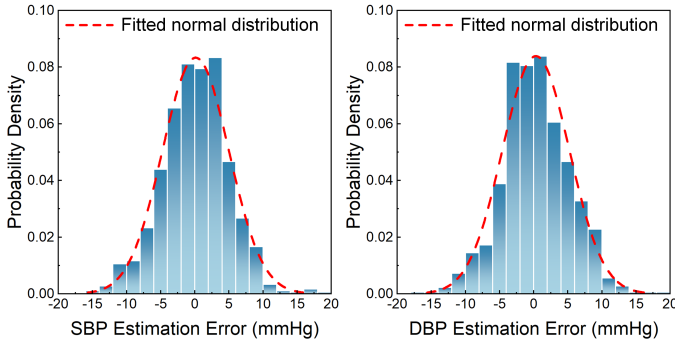


Fig. 11. Histograms of BP estimation errors based on PWV-BP with fitted normal distribution.

while those for DBP are 0.33 mmHg and -9.28 to 9.93 mmHg, respectively. Over 95% of the data points fall within the LOA range, indicating good agreement between the BP estimated by the PWV-BP and the actual BP.

We also analyze the distribution of PWV-BP estimation errors, as shown in Fig. 11. This figure presents histograms of SBP and DBP errors together with their fitted normal distribution curves. The x-axis represents the estimation errors, while the y-axis denotes the probability density. The histograms exhibit an approximately bell-shaped symmetry, with the bar heights closely matching the fitted normal distribution curves. Overall, the errors of PWV-BP follow a normal distribution, indicating stable fluctuation patterns.

TABLE II. Comparison between PWV-BP and AAMI standard.

|        |             | ME (mmHg) | STD (mmHg) |
|--------|-------------|-----------|------------|
| AAMI   | SBP and DBP | $\leq 5$  | $\leq 8$   |
| PWV-BP | SBP         | 0.13      | 4.87       |
|        | DBP         | 0.33      | 4.90       |

2) *Quantitative Assessment*: We compare the estimation errors of PWV-BP with the standard defined by the Association for the Advancement of Medical Instrumentation (AAMI) [50] in Table II. For SBP, the ME is 0.13 mmHg and the STD is 4.87 mmHg, while for DBP, the ME is 0.33 mmHg and the STD is 4.90 mmHg. These results show that the estimation

errors of SBP and DBP obtained from PWV-BP both fall within the AAMI thresholds, demonstrating the high accuracy of our proposed system.

Furthermore, we also compare the estimation errors of PWV-BP with the British Hypertension Society (BHS) [51] standard in Table III. For both SBP and DBP, more than 70% of the errors fall within 5 mmHg, over 95% within 10 mmHg, and more than 99% within 15 mmHg. Based on the BHS standard, the BP estimation performance of our proposed PWV-BP for both SBP and DBP achieves a Grade A rating, demonstrating strong consistency.

TABLE III. Comparison between PWV-BP and BHS standard.

|        |         | Cumulative Error Percentage |                   |                   |
|--------|---------|-----------------------------|-------------------|-------------------|
|        |         | $\leq 5$<br>mmHg            | $\leq 10$<br>mmHg | $\leq 15$<br>mmHg |
| BHS    | Grade C | 40%                         | 65%               | 85%               |
|        | Grade B | 50%                         | 75%               | 90%               |
|        | Grade A | 60%                         | 85%               | 95%               |
| PWV-BP | SBP     | 71.22%                      | 95.67%            | 99.56%            |
|        | DBP     | 70.00%                      | 95.29%            | 99.55%            |

### C. Comparison with the State-of-the-Art

In this section, we compare our proposed PWV-BP with three state-of-the-art methods, *i.e.*, hBP-Fi [12], AirBP [13], and mmBP [11], to provide a comprehensive performance comparison. As shown in Table IV, the comparison methods rely on pulse wave analysis from a single position to extract and analyze pulse wave signals for BP estimation. Unlike other methods, our work employs a single mmWave radar to monitor pulse waves from both the chest and the brachial artery, thereby achieving more reliable and accurate BP estimation without additional hardware requirements. The results demonstrate that our proposed PWV-BP method achieves comparable accuracy to the other methods, while offering physiological interpretability.

### D. Evaluation on the Human Arterial Model

In Section III-B1, we introduced the physiological relationship between BP and PWV and established the human arterial model. In this section, we evaluate the effectiveness of our proposed model. As illustrated in Fig. 12, the yellow line corresponds to the BP estimation model in traditional contact-based methods, which approximate PWV using the reciprocal of PTT. We find that this model yields the poorest estimation performance. This is because, across different measurements, variations in the relative position between the user and the radar device shift the position of the high quality pulse wave, thereby altering the PTD and introducing errors into the system. The orange line represents the model with PWV incorporated, which achieves a performance improvement by reducing the proportion of high-error samples and yielding fewer outliers. The blue line represents our proposed PWV-BP model incorporating BP-related features, with more than



TABLE IV. Comparison between PWV-BP and state-of-the-art methods.

| Method                         | hBP-Fi           | AirBP            | mmBP              | PWV-BP                            |
|--------------------------------|------------------|------------------|-------------------|-----------------------------------|
| Device                         | AWR1843BOOST     | BGT60TR24        | IWR1843BOOST      | AWR1443BOOST                      |
| Position                       | Arm              | Wrist            | Wrist             | Chest & Arm                       |
| BP Estimation Basis            | Waveform         | Waveform         | Waveform Features | PWV                               |
| Physiological Interpretability | Yes              | No               | No                | Yes                               |
| SBP ME $\pm$ STD (mmHg)        | $-2.05 \pm 6.83$ | $-0.30 \pm 4.80$ | $-0.87 \pm 5.01$  | <b><math>0.13 \pm 4.87</math></b> |
| DBP ME $\pm$ STD (mmHg)        | $-1.99 \pm 6.30$ | $-0.23 \pm 3.79$ | $-1.55 \pm 5.27$  | <b><math>0.33 \pm 4.90</math></b> |

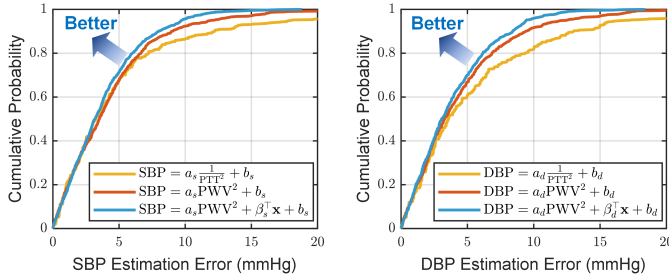


Fig. 12. Performance of different models for BP estimation.

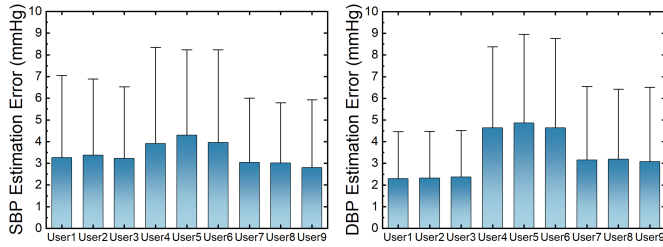


Fig. 13. BP estimation errors with different users.

70% of the estimation errors falling within 5 mmHg, thereby validating the effectiveness of our proposed human arterial model.

### E. Robustness Evaluation

1) *Robustness to User Heterogeneity*: We invited users of different ages, genders, and body weights to evaluate the accuracy of our proposed PWV-BP method. The MAEs of SBP and DBP for different users are shown in Fig. 13, where the error bars represent the absolute error range covering 90% of the data. We find that user variability exhibits no significant influence on the BP estimation performance of PWV-BP, as the MAE for all users is less than 5 mmHg. Moreover, 90% of the data fall within an absolute error of 9 mmHg.

2) *Robustness to the Number of Training Samples*: The variations in the MAE of SBP and DBP with respect to the number of training samples are shown in Fig. 14. The blue solid line demonstrates the results when the model parameters are initialized through pre-training with data from other users, while the orange dashed line denotes those when the model parameters are initialized without pre-training. The pre-trained model exhibits better estimation performance under small-sample conditions. When the number of training samples exceeds 30, the performance of the model without pre-training

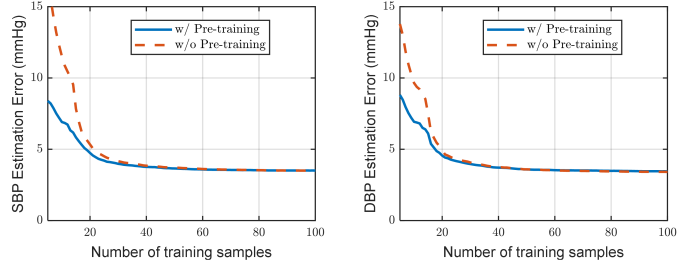


Fig. 14. Performance with different numbers of training samples.

gradually converges to that of the pre-trained model. Overall, PWV-BP reaches an MAE of 5 mmHg with 20 training samples, and the MAE continues to decrease slowly as the number of training samples increases.

### F. Strengths and Limitations

1) *Strengths*: Unlike most single-position pulse wave analysis methods, our study employs a single mmWave radar to simultaneously monitor the chest (heart) and the upper arm (brachial artery), enabling direct and synchronous measurement of PTT and PTD without multi-sensor simultaneously. This multi-position sensing improves PWV estimation fidelity and reduces sensitivity to measurement position. Moreover, by grounding BP estimation in a human arterial model, PWV-BP attains physiological interpretability absent from black-box learning methods, facilitating more reliable generalization. Finally, PWV-BP exhibits practical advantages for deployment: the two-stage candidate space construction and multi-domain filtering and fusion enhance signal quality and stability, thereby allowing the system to achieve stable accuracy with relatively few training samples. Overall, these strengths make PWV-BP a promising candidate for home-based continuous BP monitoring and early-warning applications.

2) *Limitations*: Due to individual differences in artery structures and physiological characteristics, the mapping relationship between PWV and BP is difficult to generalize across users, thereby requiring parameter calibration for new users. Notably, with our proposed human arterial model, PWV-BP can be calibrated with only about 20 samples, which is acceptable in home-use scenarios. In addition, PWV-BP also shows potential value in monitoring BP trends and variability. Future research can further explore ways to reduce or even eliminate the need for calibration, thereby enhancing system robustness.

## VI. CONCLUSION

In this paper, we have proposed a physiologically interpretable solution to contact-free BP estimation. Firstly, we have proposed a human arterial model that transforms the challenging task of BP estimation into a more tractable task of measuring PWV. Secondly, we have investigated the non-uniformity of pulse wave quality across the spatial and frequency domains, and further have analyzed the multi-domain characteristics of the pulse waves. Based on these findings, we have designed a two-stage candidate space construction strategy and a multi-domain filtering and fusion strategy, enabling the extraction of high quality pulse waves from multiple body parts. Finally, we have built a prototype and validated the system experimentally on a 77 GHz mmWave radar and have evaluated its performance under various conditions. The results have demonstrated that our proposed PWV-BP system can accurately estimate BP, achieving MEs of 0.13 and 0.33 mmHg and STDs of 4.87 and 4.90 mmHg for SBP and DBP, respectively. These outcomes confirm the system's effectiveness for BP estimation. Moreover, the multiple body parts PWV-based BP estimation strategy proposed in this paper is strongly physiologically constrained, offering new insights for future research on contact-free BP estimation.

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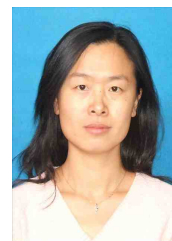
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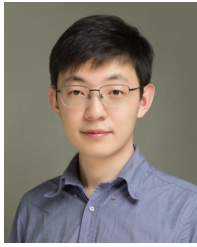


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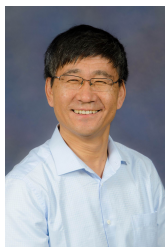
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